

1. Circle B has a diameter of 14 inches (in.). Points H and W are on the circle and form the sector HBW . The central angle $\angle HBW$ in Circle B measures 150° . Determine the area of the sector. Express your answer in terms of π .

$$A = \frac{150}{360}(\pi)(7)^2 = 20.41\bar{6}\pi \text{ in}^2$$

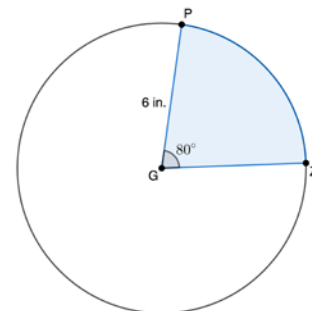
2. The area of a sector with a radius of 10 centimeters (cm) is 148.28 square cm. Calculate the approximate measure of the central angle of the sector in degrees.

$$148.28 = \frac{x}{360}(\pi)(10)^2$$

$$170^\circ = x \text{ using } \pi = 3.14$$

3. Circle G has radius $\overline{GP}$ that measures 6 inches (in.) with central angle $\angle PGZ$, where $m\angle PGZ = 80^\circ$. Find the area of the shaded sector. Round your answer to the nearest hundredth.

$$A = \frac{80}{360}(\pi)(6)^2 = 25.12 \text{ in.}^2 \text{ using } \pi = 3.14$$



4. Circle H has a radius of 9 inches (in.). Points M and N are on the circle and form the sector MHN . The central angle, $\angle MHN$, in Circle H measures 115° . Determine the area of the sector. Express your answer in terms of π .

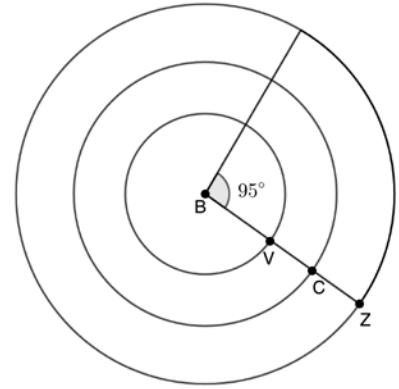
$$A = \frac{115}{360}(\pi)(9)^2 = 25.875\pi \text{ in.}^2$$

5. The area of a sector with a diameter of 12 feet (ft) is 62.8 square ft. Calculate the approximate measure of the central angle of the sector in degrees.

$$62.8 = \frac{x}{360}(\pi)(6)^2$$

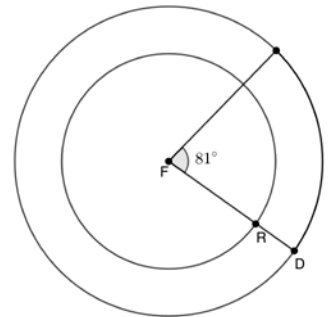
$$200^\circ = x \text{ using } \pi = 3.14$$

For questions 6-8, concentric circles with center B are shown with $m\angle B = 95^\circ$. The smallest Circle B has a radius $\overline{BV}$ that measures 12 meters (m). The largest Circle B has radius $\overline{BZ}$ and the middle Circle B has radius $\overline{BC}$. Each circle has a radius that is 12 m longer than the immediate prior circle.



6. What is the arc length of the circle with radius $\overline{BV}$?
 $\frac{95}{360}(2\pi)(12) \approx 19.89 \text{ m. using } \pi = 3.14$
7. What is the arc length of the circle with radius $\overline{BC}$?
 $\frac{95}{360}(2\pi)(24) \approx 39.77 \text{ m. using } \pi = 3.14$
8. What is the arc length of the circle with radius $\overline{BZ}$?
 $\frac{95}{360}(2\pi)(36) \approx 59.66 \text{ m. using } \pi = 3.14$

For questions 9-10, concentric circles with center F are shown with $m\angle F = 81^\circ$. The smallest Circle F has a radius $\overline{FR}$ that measures 15 meters (m) and the largest Circle F has radius $\overline{FD}$. The radius of the largest Circle F has a radius 7 m larger than the smaller Circle F .



9. What is the arc length of the circle with radius $\overline{FR}$?
 $\frac{81}{360}(2\pi)(15) \approx 21.195 \text{ m. using } \pi = 3.14$
10. What is the arc length of the circle with radius $\overline{FD}$?
 $\frac{81}{360}(2\pi)(22) \approx 31.086 \text{ m. using } \pi = 3.14$